

31.1 Generalized least squares

The model of this chapter keeps the mean structure of every earlier chapter and loosens the second moments:

$$E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}, \quad \text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{V}, \quad (31.1.1)$$

where $\mathbf{X}$ is $n \times p$ of rank r , the matrix $\mathbf{V}$ is known and nonnegative definite, and $\sigma^2 > 0$ is unknown. This section and the next assume $\mathbf{V}$ nonsingular; [Section 31.3](#) removes that assumption.

31.1.1 What is assumed and what is not

The split of the covariance into a known matrix $\mathbf{V}$ and an unknown scalar σ^2 is a convention, not extra knowledge. Only the product $\sigma^2\mathbf{V}$ enters [Eq. 31.1.1](#), so $(\sigma^2, \mathbf{V})$ and $(\sigma^2/c, c\mathbf{V})$ describe the same model for every $c > 0$; it is usual to normalize $\mathbf{V}$ so that $\text{tr}(\mathbf{V})/n = 1$, making σ^2 the variance of a single observation on a reference scale. What the model really assumes is that the *shape* of the covariance — the ratios of variances and all the correlations — is known.

That is exactly true only when the design forces the shape, as when an observation is the mean of m_i replicates and has variance σ^2/m_i ([Section 21.3](#)), or when the construction of the data does, as under the linear transformations of [Section 31.3](#). Otherwise $\mathbf{V}$ belongs to a family $\mathbf{V}(\boldsymbol{\theta})$ estimated first and then treated as known, and this section's exactness is a fiction whose price is computed in [Section 31.4](#).

31.1.2 The estimator and the whitened model

Generalized least squares was defined in [Definition 6.9.1](#) as projection in the $\mathbf{V}^{-1}$ inner product; we repeat it in the notation of this part.

Definition 31.1.1. Generalized least squares

Under [Eq. 31.1.1](#) with $\mathbf{V}$ positive definite, a **generalized least squares estimate** is any $\hat{\boldsymbol{\beta}}_{\mathbf{V}}$ minimizing

$$Q_{\mathbf{V}}(\mathbf{b}) = (\mathbf{y} - \mathbf{X}\mathbf{b})^{\top} \mathbf{V}^{-1} (\mathbf{y} - \mathbf{X}\mathbf{b}),$$

equivalently any solution of the **generalized normal equations** $\mathbf{X}^{\top} \mathbf{V}^{-1} \mathbf{X}\mathbf{b} = \mathbf{X}^{\top} \mathbf{V}^{-1} \mathbf{y}$. The fitted vector is $\mathbf{X}\hat{\boldsymbol{\beta}}_{\mathbf{V}} = \mathbf{A}_{\mathbf{V}}\mathbf{y}$, where

$$\mathbf{A}_{\mathbf{V}} = \mathbf{X}(\mathbf{X}^{\top} \mathbf{V}^{-1} \mathbf{X})^{-} \mathbf{X}^{\top} \mathbf{V}^{-1} \quad (31.1.2)$$

for any generalized inverse, and the **generalized residual** is $\hat{\boldsymbol{\varepsilon}}_{\mathbf{V}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{\mathbf{V}}$.

Everything algebraic about $\mathbf{A}_{\mathbf{V}}$ is [Theorem 6.9.1](#) with $\mathbf{A} = \mathbf{V}^{-1}$: the choice of generalized inverse is immaterial, it is idempotent with column space $\mathcal{C}(\mathbf{X})$, and $\mathbf{V}^{-1}\mathbf{A}_{\mathbf{V}}$ is symmetric although $\mathbf{A}_{\mathbf{V}}$ is not — in the Euclidean geometry it is an oblique projection ([Proposition 6.9.4](#)). Everything statistical comes from one change of variables. Let $\mathbf{V}^{-1/2}$ be the symmetric positive definite square root of $\mathbf{V}^{-1}$ ([Theorem 1.7.6](#)) and set

$$\mathbf{Y}_* = \mathbf{V}^{-1/2}\mathbf{Y}, \quad \mathbf{X}_* = \mathbf{V}^{-1/2}\mathbf{X}. \quad (31.1.3)$$

By Theorem 2.2.1, $E(\mathbf{Y}_*) = \mathbf{X}_*\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y}_*) = \sigma^2\mathbf{I}$: the **whitened model** satisfies the assumptions of Part II exactly. Since $Q_{\mathbf{V}}(\mathbf{b}) = \|\mathbf{Y}_* - \mathbf{X}_*\mathbf{b}\|^2$, generalized least squares for $(\mathbf{Y}, \mathbf{X})$ is ordinary least squares for $(\mathbf{Y}_*, \mathbf{X}_*)$. Because $\mathbf{V}^{-1/2}$ is nonsingular, $\text{rank}(\mathbf{X}_*) = r$ and $\mathcal{C}(\mathbf{X}_*^\top) = \mathcal{C}(\mathbf{X}^\top)$, so by Theorem 8.3.1 the two models have the same estimable functions: nothing about identifiability changes. Any nonsingular $\mathbf{L}$ with $\mathbf{V} = \mathbf{L}\mathbf{L}^\top$ serves equally well, by Proposition 2.2.4(b); the Cholesky factor is the usual choice in computation.

Whitening is rarely carried out by forming $\mathbf{V}^{-1/2}$; structured $\mathbf{V}$ almost always has a shortcut. If the observations fall into G clusters of size m that share a random shock, so that $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho\mathbf{Z}\mathbf{Z}^\top$ with $\mathbf{Z}$ the $n \times G$ matrix of cluster indicators, then Theorem 1.6.5 gives

$$\mathbf{V}^{-1} = \frac{1}{1 - \rho} \left(\mathbf{I} - \frac{\rho}{1 + (m - 1)\rho} \mathbf{Z}\mathbf{Z}^\top \right), \quad (31.1.4)$$

and the whitening is the **quasi-demeaning**

$$y_{kj} \mapsto \frac{y_{kj} - c\bar{y}_k}{\sqrt{1 - \rho}}, \quad c = 1 - \sqrt{\frac{1 - \rho}{1 + (m - 1)\rho}}, \quad (31.1.5)$$

where $\bar{y}_k$ is the mean of cluster k (Exercise (B1)). At $\rho = 0$, $c = 0$ and nothing happens; at $\rho = 0.5$ with $m = 12$, $c = 0.723$; as $\rho \rightarrow 1$, $c \rightarrow 1$ and each cluster is fully centred, throwing away everything the shared shock contaminated. Generalized least squares interpolates, and Chapter 32 will recognize Eq. 31.1.5 as the transformation a random intercept induces.

31.1.3 Exact inference when the shape is known

Theorem 31.1.1. Inference under a known covariance shape

Let $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{V})$ with $\mathbf{V}$ known and positive definite, $r = \text{rank}(\mathbf{X}) < n$. Put

$$\text{SSE}_{\mathbf{V}} = \hat{\boldsymbol{\varepsilon}}_{\mathbf{V}}^\top \mathbf{V}^{-1} \hat{\boldsymbol{\varepsilon}}_{\mathbf{V}} = \mathbf{Y}^\top \mathbf{V}^{-1} (\mathbf{I} - \mathbf{A}_{\mathbf{V}}) \mathbf{Y}, \quad s_{\mathbf{V}}^2 = \frac{\text{SSE}_{\mathbf{V}}}{n - r}. \quad (31.1.6)$$

- (a) For $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$, the statistic $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}_{\mathbf{V}}$ is the same for every solution of the generalized normal equations, and

$$\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}_{\mathbf{V}} \sim N(\boldsymbol{\lambda}^\top \boldsymbol{\beta}, \sigma^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \boldsymbol{\lambda}).$$

It is the best linear unbiased estimator of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ (Corollary 7.2.3).

- (b) $\text{SSE}_{\mathbf{V}}/\sigma^2 \sim \chi^2(n - r)$, independently of $\mathbf{X}\hat{\boldsymbol{\beta}}_{\mathbf{V}}$.
 (c) $E(s_{\mathbf{V}}^2) = \sigma^2$ and $\text{Var}(s_{\mathbf{V}}^2) = 2\sigma^4/(n - r)$.
 (d) For estimable $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ with $\boldsymbol{\lambda} \neq \mathbf{0}$,

$$\frac{\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}_{\mathbf{V}} - \boldsymbol{\lambda}^\top \boldsymbol{\beta}}{s_{\mathbf{V}} \sqrt{\boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \boldsymbol{\lambda}}} \sim t(n - r).$$

- (e) Let $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$ with $\text{rank}(\mathbf{X}_0) = r_0 < r$, and let $\text{SSE}_{\mathbf{V},0}$ be Eq. 31.1.6 computed in the reduced model. Then

$$F = \frac{(\text{SSE}_{\mathbf{V},0} - \text{SSE}_{\mathbf{V}})/(r - r_0)}{s_{\mathbf{V}}^2} \sim F(r - r_0, n - r, \gamma), \quad (31.1.7)$$

with $\gamma = \|(\mathbf{M}_* - \mathbf{M}_{*0})\mathbf{X}_*\boldsymbol{\beta}\|^2/\sigma^2$, where $\mathbf{M}_*$ and $\mathbf{M}_{*0}$ are the orthogonal projections onto $\mathcal{C}(\mathbf{V}^{-1/2}\mathbf{X})$ and $\mathcal{C}(\mathbf{V}^{-1/2}\mathbf{X}_0)$. The noncentrality is zero iff $\mathbf{X}_*\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}_0)$.

Proof. Pass to the whitened model Eq. 31.1.3, which satisfies the normal linear model Eq. 7.3.1 with model matrix $\mathbf{X}_*$ of rank r . As shown above, $\hat{\boldsymbol{\beta}}_{\mathbf{V}}$ is exactly a least squares estimate for $(\mathbf{Y}_*, \mathbf{X}_*)$, and the two models have the same estimable functions. Moreover

$$\text{SSE}_{\mathbf{V}} = \|\mathbf{Y}_* - \mathbf{X}_*\hat{\boldsymbol{\beta}}_{\mathbf{V}}\|^2 = \|(\mathbf{I} - \mathbf{M}_*)\mathbf{Y}_*\|^2$$

is the residual sum of squares of the whitened problem, because $\mathbf{V}^{-1/2}\hat{\boldsymbol{\varepsilon}}_{\mathbf{V}} = (\mathbf{I} - \mathbf{M}_*)\mathbf{Y}_*$ by Theorem 6.4.1, and $\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} = \mathbf{X}_*^\top \mathbf{X}_*$.

Now apply Theorem 7.5.1 to $(\mathbf{Y}_*, \mathbf{X}_*)$. Part (a) of that theorem gives (a) here, since $\boldsymbol{\lambda}^\top (\mathbf{X}_*^\top \mathbf{X}_*)^{-1} \boldsymbol{\lambda} = \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \boldsymbol{\lambda}$ and invariance of the estimate follows from Theorem 6.4.4 applied in the whitened model. Parts (b) and (c) of Theorem 7.5.1 give (b) here, noting that $\mathbf{X}_*\hat{\boldsymbol{\beta}}_{\mathbf{V}} = \mathbf{V}^{1/2}\mathbf{M}_*\mathbf{Y}_*$ is a function of $\mathbf{M}_*\mathbf{Y}_*$, and part (d) gives (c). Part (d) follows from (a) and (b) exactly as Theorem 12.1.1 does in the Euclidean case. For (e), $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$ implies $\mathcal{C}(\mathbf{V}^{-1/2}\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{V}^{-1/2}\mathbf{X})$ with ranks $r_0 < r$, so Definition 11.1.1 holds for the whitened pair and Theorem 11.1.1 applies; $\text{SSE}_{\mathbf{V},0} = \|(\mathbf{I} - \mathbf{M}_{*0})\mathbf{Y}_*\|^2$ by the same computation. The noncentrality Eq. 11.1.2 is $\|(\mathbf{M}_* - \mathbf{M}_{*0})\mathbf{X}_*\boldsymbol{\beta}\|^2/\sigma^2$, and it vanishes iff $\mathbf{X}_*\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}_{*0})$, that is, iff $\mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}_0)$ since $\mathbf{V}^{-1/2}$ is nonsingular. ■

The theorem is not a new theory: it is Parts II and III read in another coordinate system, and the proof is nothing but the dictionary. Under normality $\hat{\boldsymbol{\beta}}_{\mathbf{V}}$ also maximizes the likelihood (Exercise (B3)), and $s_{\mathbf{V}}^2$ has smallest variance among unbiased estimates of σ^2 , by Section 7.4 applied to the whitened model.

31.1.4 What changes, and what does not

The dictionary is exact, but several habits from Parts II and III do not survive translation.

Warning. Under Eq. 31.1.1 with $\mathbf{V} \neq \mathbf{I}$:

- The generalized residual is **not** orthogonal to $\mathcal{C}(\mathbf{X})$; what holds is $\mathbf{X}^\top \mathbf{V}^{-1} \hat{\boldsymbol{\varepsilon}}_{\mathbf{V}} = \mathbf{0}$. The entries of $\hat{\boldsymbol{\varepsilon}}_{\mathbf{V}}$ need not sum to zero even when the model has an intercept.
- $\text{SSE}_{\mathbf{V}}$ is **not** $\sum_i (y_i - \hat{y}_i)^2$, and the ordinary residual sum of squares of the generalized fit is necessarily *larger* than that of the ordinary fit, which minimizes it. A model comparison using $\sum_i (y_i - \hat{y}_i)^2$ always prefers ordinary least squares and proves nothing.
- R^2 has no agreed meaning: the squared-cosine reading of Theorem 6.7.2 holds in the $\mathbf{V}^{-1}$ geometry, not the Euclidean one.
- The diagonal entries of $\mathbf{A}_{\mathbf{V}}$ still sum to r , but they are not leverages in the sense of Proposition 6.8.1 and can fall outside $[0, 1]$ (Exercise (B2)).

What survives is everything inference needs: unbiasedness, optimality, the χ^2 , t and F distribu-

tions, the degrees of freedom $n - r$, and γ as a squared distance from the reduced model space, measured now in the $\mathbf{V}^{-1}$ metric.

Example. A trend in sea surface temperature

The El Niño data record the monthly averaged sea surface temperature of a region of the eastern Pacific for the 61 complete years 1950 to 2010, 732 observations in 61 blocks of 12. Fit a linear trend in time together with three harmonic pairs at one, two and three cycles per year, so $p = 8$. Warm and cold years are warm and cold in every month, so the natural working model is a random annual shock: equicorrelation ρ inside each year, independence across years. Take $\rho = 0.5$ as known for the moment.

Ordinary least squares gives a trend of 0.1349 degrees per decade with the usual standard error 0.0223; generalized least squares gives 0.1347 with standard error 0.0512, larger by a factor of 2.30. The estimates agree to three decimals, the standard errors not at all. The annual cycle tells the same story in reverse: the coefficient of $\sin(2\pi \text{ month}/12)$ is 2.7629 under both fits, with standard error 0.0554 from ordinary and 0.0354 from generalized least squares — this time the usual standard error is too *large*. [Section 31.2](#) explains both facts.

The F test of “no trend” is 36.7 if the annual shock is ignored and 6.92 once it is allowed for, on 1 and 724 degrees of freedom ($p = 0.0087$): still clear evidence, but with a tenth of the apparent weight. The test that the second and third harmonics are absent moves the other way, from 10.4 to 25.6, because the annual shock inflates s^2 and so deflates every F whose numerator lives inside years. [Figure 31.1.1](#)(a) shows the annual mean residuals, far too spread out for independent errors, and panel (b) the two confidence bands.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

frame = sm.datasets.elnino.load_pandas().data
years = frame["YEAR"].to_numpy()
temps = frame.iloc[:, 1:].to_numpy()           # 61 years by 12 months
G, m = temps.shape
n = G * m
y = temps.ravel()                             # the month varies fastest
year = np.repeat(years, m)
month = np.tile(np.arange(1, m + 1), G)
time = (year - 1980.0) + (month - 6.5) / 12.0   # decimal years, centred at
1980

cols = [np.ones(n), time / 10.0]               # the trend is per decade
for k in (1, 2, 3):
    cols += [np.cos(2 * np.pi * k * month / m), np.sin(2 * np.pi * k * month / m)]
X = np.column_stack(cols)
p = X.shape[1]
```

[Run this code online](#)

```
rho = 0.5                                     # taken as known for now

def gls_fit(X, y, rho, m):
    """GLS with equicorrelation rho inside blocks of m consecutive observations.
```

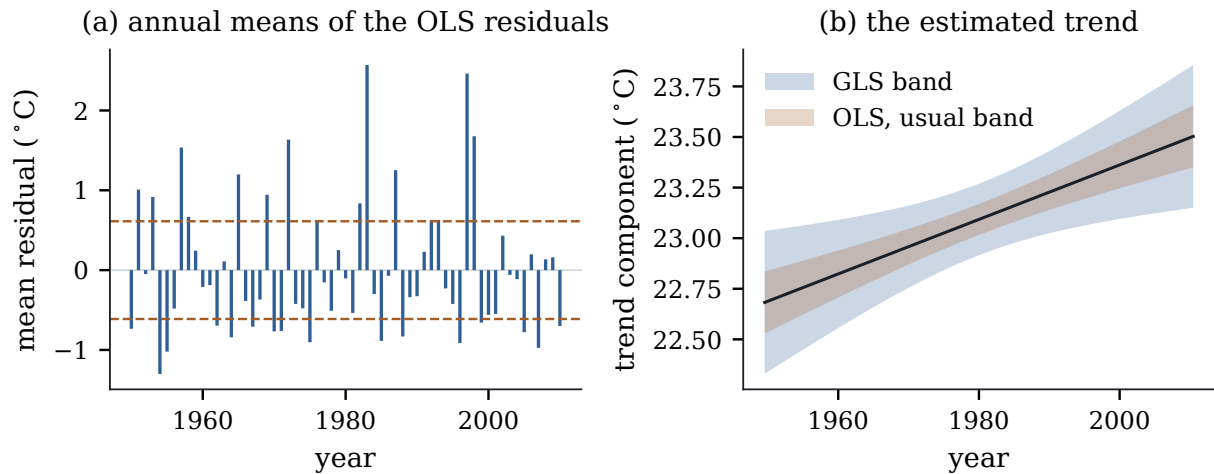


Figure 31.1.1: El Niño sea surface temperatures, 1950–2010. (a) The annual means of the ordinary least squares residuals, with the ± 2 standard deviation lines that independent errors would give (dashed). Whole years are warm or cold together. (b) The estimated trend component with nominal 95% bands, from ordinary least squares with the usual covariance and from generalized least squares with $\rho = 0.5$.

```

Whitening is the quasi-demeaning  $y \rightarrow (y - c \bar{y}) / \sqrt{1 - \rho}$  with
 $c = 1 - \sqrt{(1 - \rho) / (1 + (m - 1) \rho)}$ ; ordinary least squares on the
whitened
data is generalized least squares on the original data.
"""
c = 1.0 - np.sqrt((1 - rho) / (1 + (m - 1) * rho))

def whiten(a):
    b = a.reshape(len(a) // m, m, -1) if a.ndim > 1 else a.reshape(-1, m)
    return ((b - c * b.mean(axis=1, keepdims=True)) / np.sqrt(1 - rho)).reshape(
a.shape)

Xw, yw = whiten(X), whiten(y)
b = np.linalg.solve(Xw.T @ Xw, Xw.T @ yw)
rss = float((yw - Xw @ b) @ (yw - Xw @ b))           # the  $V^{-1}$  residual sum of
squares
s2 = rss / (len(y) - X.shape[1])
return b, s2 * np.linalg.inv(Xw.T @ Xw), rss, c

b_ols = np.linalg.solve(X.T @ X, X.T @ y)
rss_ols = float((y - X @ b_ols) @ (y - X @ b_ols))
V_ols = rss_ols / (n - p) * np.linalg.inv(X.T @ X)   # the standard errors usually
reported
b_gls, V_gls, rss_gls, cq = gls_fit(X, y, rho, m)

print(f"trend per decade: OLS {b_ols[1]:+.4f} GLS {b_gls[1]:+.4f}")
print(f"standard error: OLS {np.sqrt(V_ols[1, 1]):.4f} GLS {np.sqrt(V_gls[1, 1]):.4f}")
print(f"quasi-demeaning constant c = {cq:.4f}, s_GLS = {np.sqrt(rss_gls / (n - p)):.4f}")

```

[Run this code online](#)

The F test of [Theorem 31.1.1\(e\)](#) is computed exactly as in Part III: fit the full and the reduced

model in the same geometry and compare their $\mathbf{V}^{-1}$ residual sums of squares. The listing does this for both hypotheses of [Example \(A trend in sea surface temperature\)](#), using $\rho = 0$ to recover the ordinary analysis.

```
def f_test(X, y, drop, rho, m):
    """F test that the coefficients listed in `drop` are zero, in the rho-geometry.
    """
    keep = [j for j in range(X.shape[1]) if j not in drop]
    full, null = gls_fit(X, y, rho, m), gls_fit(X[:, keep], y, rho, m)
    df1, df2 = len(drop), len(y) - X.shape[1]
    F = (null[2] - full[2]) / df1 / (full[2] / df2)
    return F, stats.f.sf(F, df1, df2)

F_trend_gls, p_trend_gls = f_test(X, y, [1], rho, m)
F_trend_ols, p_trend_ols = f_test(X, y, [1], 0.0, m)  # rho = 0 is ordinary least
               squares
F_harm_gls, p_harm_gls = f_test(X, y, [4, 5, 6, 7], rho, m)
F_harm_ols, p_harm_ols = f_test(X, y, [4, 5, 6, 7], 0.0, m)
print(f"trend:      F = {F_trend_ols:7.2f} (rho = 0)    F = {F_trend_gls:6.2f} (rho =
      0.5)")
print(f"harmonics:  F = {F_harm_ols:7.2f} (rho = 0)    F = {F_harm_gls:6.2f} (rho =
      0.5)")
```

[Run this code online](#)

Idea. Generalized least squares is one change of coordinates away from ordinary least squares, and everything in Parts II and III transfers — provided *all* of it does: the estimate, the residual sum of squares, the degrees of freedom and the sums of squares of every hypothesis. Mixing geometries, as in a generalized estimate with an ordinary standard error, is the one way to get this wrong.

31.1.5 Exercises

31.1.5.1 A. Check your understanding

Exercise. A1

Show that replacing $(\sigma^2, \mathbf{V})$ by $(\sigma^2/c, c\mathbf{V})$ for $c > 0$ leaves $\hat{\boldsymbol{\beta}}_{\mathbf{V}}$, $\mathbf{A}_{\mathbf{V}}$ and every statistic in [Theorem 31.1.1](#)(d), (e) unchanged, and multiplies $\text{SSE}_{\mathbf{V}}$ and $s_{\mathbf{V}}^2$ by $1/c$. Conclude that no normalization of $\mathbf{V}$ affects any inference about $\boldsymbol{\beta}$.

Solution. The criterion $Q_{c\mathbf{V}}(\mathbf{b}) = Q_{\mathbf{V}}(\mathbf{b})/c$ is minimized at the same $\mathbf{b}$, and $(c\mathbf{V})^{-1} = \mathbf{V}^{-1}/c$ cancels from [Eq. 31.1.2](#). Then $\text{SSE}_{c\mathbf{V}} = \text{SSE}_{\mathbf{V}}/c$ and $\boldsymbol{\lambda}^{\top}(\mathbf{X}^{\top}(c\mathbf{V})^{-1}\mathbf{X})^{-1}\boldsymbol{\lambda} = c\boldsymbol{\lambda}^{\top}(\mathbf{X}^{\top}\mathbf{V}^{-1}\mathbf{X})^{-1}\boldsymbol{\lambda}$, so the factors c cancel in the t statistic; in F they cancel between numerator and denominator.

Exercise. A2

Show that $\mathbf{1}^\top \hat{\boldsymbol{\varepsilon}}_{\mathbf{V}} = 0$ for every $\mathbf{y}$ iff $\mathbf{1} \in \mathcal{C}(\mathbf{V}^{-1}\mathbf{X})$. Give an example with an intercept in which this fails.

31.1.5.2 B. Practice

Exercise. B1

Let $\mathbf{V}_0 = (1 - \rho)\mathbf{I}_m + \rho\mathbf{1}\mathbf{1}^\top$ with $-1/(m - 1) < \rho < 1$. Verify Eq. 31.1.4 by Theorem 1.6.5, and show that $\mathbf{L}_0 = (\mathbf{I}_m - c\mathbf{1}\mathbf{1}^\top/m)/\sqrt{1 - \rho}$ satisfies $\mathbf{L}_0\mathbf{V}_0\mathbf{L}_0^\top = \mathbf{I}_m$ for $c = 1 - \sqrt{(1 - \rho)/(1 + (m - 1)\rho)}$. Deduce Eq. 31.1.5.

Solution. Write $\mathbf{P} = \mathbf{1}\mathbf{1}^\top/m$, which is idempotent, so $\mathbf{V}_0 = (1 - \rho)\mathbf{I} + \rho m\mathbf{P}$. For any c , $(\mathbf{I} - c\mathbf{P})\mathbf{V}_0(\mathbf{I} - c\mathbf{P}) = (1 - \rho)\mathbf{I} + [(1 - \rho)(c^2 - 2c) + \rho m(1 - c)^2]\mathbf{P}$, using $\mathbf{P}^2 = \mathbf{P}$. Setting $a = 1 - \rho$ and $b = \rho m$, the bracket vanishes iff $(a + b)c^2 - 2(a + b)c + b = 0$, that is $c^2 - 2c + b/(a + b) = 0$, whose root in $[0, 1]$ is $c = 1 - \sqrt{a/(a + b)}$. Dividing by $1 - \rho$ gives $\mathbf{L}_0\mathbf{V}_0\mathbf{L}_0^\top = \mathbf{I}$. The Woodbury identity with $\mathbf{U} = \mathbf{1}$ gives Eq. 31.1.4 directly. Applying $\mathbf{L}_0$ block by block is Eq. 31.1.5, since $\mathbf{1}\mathbf{1}^\top \mathbf{y}_k/m = \bar{y}_k\mathbf{1}$.

Exercise. B2

Show that $\text{tr } \mathbf{A}_{\mathbf{V}} = r$ for every positive definite $\mathbf{V}$. Then take $n = 2$, $p = 1$,

$$\mathbf{X} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \mathbf{V} = \begin{pmatrix} 1 & 0.9 \\ 0.9 & 1 \end{pmatrix},$$

and show that the diagonal of $\mathbf{A}_{\mathbf{V}}$ is $(-4/7, 11/7)$. Why does this not contradict Proposition 6.8.1?

Exercise. B3

Show that $\hat{\boldsymbol{\varepsilon}}_{\mathbf{V}}^\top \hat{\boldsymbol{\varepsilon}}_{\mathbf{V}} \geq \hat{\boldsymbol{\varepsilon}}^\top \hat{\boldsymbol{\varepsilon}}$, where $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{y}$ is the ordinary residual, with equality iff $\mathbf{X}\hat{\boldsymbol{\beta}}_{\mathbf{V}} = \mathbf{M}\mathbf{y}$. Show also that $\hat{\boldsymbol{\varepsilon}}^\top \mathbf{V}^{-1} \hat{\boldsymbol{\varepsilon}} \geq \hat{\boldsymbol{\varepsilon}}_{\mathbf{V}}^\top \mathbf{V}^{-1} \hat{\boldsymbol{\varepsilon}}_{\mathbf{V}}$. Explain why neither inequality says anything about which fit is better.

Exercise. B4

Observations Y_{kj} , $j = 1, \dots, m_k$, $k = 1, \dots, G$, have mean μ and covariance $\sigma^2\mathbf{V}$ with $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho\mathbf{Z}\mathbf{Z}^\top$, $\mathbf{Z}$ the cluster indicator matrix. Using Eq. 31.1.4 cluster by cluster, show that

$$\hat{\mu}_{\mathbf{V}} = \frac{\sum_k w_k \bar{Y}_k}{\sum_k w_k}, \quad w_k = \frac{m_k}{1 - \rho + m_k \rho}, \quad (31.1.8)$$

with $\text{Var}(\hat{\mu}_{\mathbf{V}}) = \sigma^2 / \sum_k w_k$. Interpret the weights as $\rho \rightarrow 0$ and $\rho \rightarrow 1$.

Solution. With $\mathbf{V}$ block diagonal, $\mathbf{V}^{-1}$ is block diagonal with blocks $(\mathbf{I} - \frac{\rho}{1-\rho+m_k\rho}\mathbf{1}\mathbf{1}^\top)/(1-\rho)$. Hence $\mathbf{1}^\top \mathbf{V}^{-1} \mathbf{1} = \sum_k (m_k - \rho m_k^2 / (1 - \rho + m_k \rho)) / (1 - \rho) = \sum_k w_k$, and $\mathbf{1}^\top \mathbf{V}^{-1} \mathbf{y} = \sum_k w_k \bar{y}_k$ by the same computation. The generalized normal equation is $(\mathbf{1}^\top \mathbf{V}^{-1} \mathbf{1})\mu = \mathbf{1}^\top \mathbf{V}^{-1} \mathbf{y}$. As $\rho \rightarrow 0$, $w_k \rightarrow m_k$ and $\hat{\mu}_{\mathbf{V}} \rightarrow \bar{y}$; as $\rho \rightarrow 1$, $w_k \rightarrow 1$ and every cluster counts once, however large, because the extra observations only repeat the same shock.

31.1.5.3 C. Going deeper

Exercise. C1

Suppose $(\mathbf{Y}^\top, Y_0)^\top$ is jointly distributed with $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$, $E(Y_0) = \mathbf{x}_0^\top \boldsymbol{\beta}$, $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}$, $\text{Cov}(\mathbf{Y}, Y_0) = \sigma^2 \mathbf{c}$ and $\text{Var}(Y_0) = \sigma^2 v_0$, with $\mathbf{V}$ positive definite and $\text{rank}(\mathbf{X}) = p$. Among predictors $\mathbf{a}^\top \mathbf{Y}$ with $E(\mathbf{a}^\top \mathbf{Y} - Y_0) = 0$ for all $\boldsymbol{\beta}$, show that the one minimizing $E(\mathbf{a}^\top \mathbf{Y} - Y_0)^2$ is

$$\hat{Y}_0 = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}_{\mathbf{V}} + \mathbf{c}^\top \mathbf{V}^{-1}(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{\mathbf{V}}),$$

with prediction error variance $\sigma^2[v_0 - \mathbf{c}^\top \mathbf{V}^{-1} \mathbf{c} + \mathbf{d}^\top (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{d}]$, $\mathbf{d} = \mathbf{x}_0 - \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{c}$. This is Goldberger's formula; [Chapter 32](#) recognizes it as the best linear unbiased predictor of a random effect.

Solution. Unbiasedness requires $\mathbf{X}^\top \mathbf{a} = \mathbf{x}_0$, and $E(\mathbf{a}^\top \mathbf{Y} - Y_0)^2 = \sigma^2(\mathbf{a}^\top \mathbf{V} \mathbf{a} - 2\mathbf{a}^\top \mathbf{c} + v_0)$. By Proposition 1.11.3 with the positive definite $\mathbf{V}$, the minimizer is $\mathbf{a} = \mathbf{V}^{-1} \mathbf{c} + \mathbf{V}^{-1} \mathbf{X}(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{d}$, and collecting terms gives $\mathbf{a}^\top \mathbf{Y} = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}_{\mathbf{V}} + \mathbf{c}^\top \mathbf{V}^{-1}(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{\mathbf{V}})$; the minimum value is the stated variance. Setting $\mathbf{c} = \mathbf{0}$ recovers the ordinary prediction of [Chapter 12](#). A correlated future observation is predicted by correcting the fitted mean with the part of the current residual that predicts it.

Exercise. C2

Let $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{V}(\boldsymbol{\theta}))$ with $\text{rank}(\mathbf{X}) = p$. Show that for each fixed $\boldsymbol{\theta}$ the log-likelihood is maximized over $(\boldsymbol{\beta}, \sigma^2)$ at $\hat{\boldsymbol{\beta}}_{\mathbf{V}(\boldsymbol{\theta})}$ and $\hat{\sigma}^2(\boldsymbol{\theta}) = \text{SSE}_{\mathbf{V}(\boldsymbol{\theta})}/n$, and that the profile log-likelihood is

$$\ell_p(\boldsymbol{\theta}) = -\frac{n}{2} \log \hat{\sigma}^2(\boldsymbol{\theta}) - \frac{1}{2} \log \det \mathbf{V}(\boldsymbol{\theta}) + \text{const.}$$

Why does the determinant term make the normalization of $\mathbf{V}$ matter here, when [Exercise \(A1\)](#) says it does not matter for inference about $\boldsymbol{\beta}$?